

Task 2: Data Visualization and Storytelling

Superstore Sales Analysis

This project analyzes the Superstore sales dataset using Python, Pandas, and Matplotlib to identify sales patterns, profitability, regional performance, and business insights.

Tools: Python, Pandas, Matplotlib, Google Colab

Dataset: Superstore Sales Dataset from Kaggle

```
from google.colab import files
```

```
uploaded = files.upload()
```

[Choose Files](#) SampleSuperstore.csv

SampleSuperstore.csv(text/csv) - 1113007 bytes, last modified: 10/2/2026 - 100% done
Saving SampleSuperstore.csv to SampleSuperstore.csv

```
import pandas as pd
```

```
filename = list(uploaded.keys())[0]
```

```
df = pd.read_csv(filename, encoding='latin-1')
```

```
print("Dataset loaded successfully!")
```

```
print("Rows:", df.shape[0])
```

```
print("Columns:", df.shape[1])
```

```
df.head()
```

Dataset loaded successfully!

Rows: 9994

Columns: 13

	Ship Mode	Segment	Country	City	State	Postal Code	Region	Category
0	Second Class	Consumer	United States	Henderson	Kentucky	42420	South	Furniture
1	Second Class	Consumer	United States	Henderson	Kentucky	42420	South	Furniture
2	Second Class	Corporate	United States	Los Angeles	California	90036	West	Office Supplies

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```
print(df.columns.tolist())
```

```
['Ship Mode', 'Segment', 'Country', 'City', 'State', 'Postal Code', 'Region',
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 13 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Ship Mode       9994 non-null   object
1   Segment         9994 non-null   object
2   Country         9994 non-null   object
3   City            9994 non-null   object
4   State           9994 non-null   object
5   Postal Code     9994 non-null   int64
6   Region          9994 non-null   object
7   Category        9994 non-null   object
8   Sub-Category    9994 non-null   object
9   Sales           9994 non-null   float64
10  Quantity        9994 non-null   int64
11  Discount         9994 non-null   float64
12  Profit          9994 non-null   float64
dtypes: float64(3), int64(2), object(8)
memory usage: 1015.1+ KB
```

```
df.isnull().sum()
```

	0
Ship Mode	0
Segment	0
Country	0
City	0
State	0
Postal Code	0
Region	0
Category	0
Sub-Category	0
Sales	0
Quantity	0
Discount	0
Profit	0

```
dtype: int64
```

```
print("Duplicate rows:", df.duplicated().sum())
```

Duplicate rows: 17

```
df.columns = df.columns.str.strip()
```

```
print(df.columns.tolist())
```

```
['Ship Mode', 'Segment', 'Country', 'City', 'State', 'Postal Code', 'Region',
```

```
df.columns = df.columns.str.strip()
```

```
df['Sales'] = pd.to_numeric(df['Sales'], errors='coerce')
```

```
df['Profit'] = pd.to_numeric(df['Profit'], errors='coerce')
```

```
df['Discount'] = pd.to_numeric(df['Discount'], errors='coerce')
```

```
df['Quantity'] = pd.to_numeric(df['Quantity'], errors='coerce')
```

```
print("Data preparation completed!")
```

```
print("Rows:", df.shape[0])
```

```
print("Columns:", df.shape[1])
```

```
df.head()
```

Data preparation completed!

Rows: 9994

Columns: 13

	Ship Mode	Segment	Country	City	State	Postal Code	Region	Category
0	Second Class	Consumer	United States	Henderson	Kentucky	42420	South	Furniture
1	Second Class	Consumer	United States	Henderson	Kentucky	42420	South	Furniture
2	Second Class	Corporate	United States	Los Angeles	California	90036	West	Office Supplies

Next steps:

[Generate code with df](#)
[New interactive sheet](#)

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
```

```
plt.rcParams['figure.figsize'] = (10, 6)
```

```
print("Visualization libraries loaded!")
```

Visualization libraries loaded!

✓ Sales by Product Category

```
category_sales = df.groupby('Category')['Sales'].sum().sort_values(ascending=False)

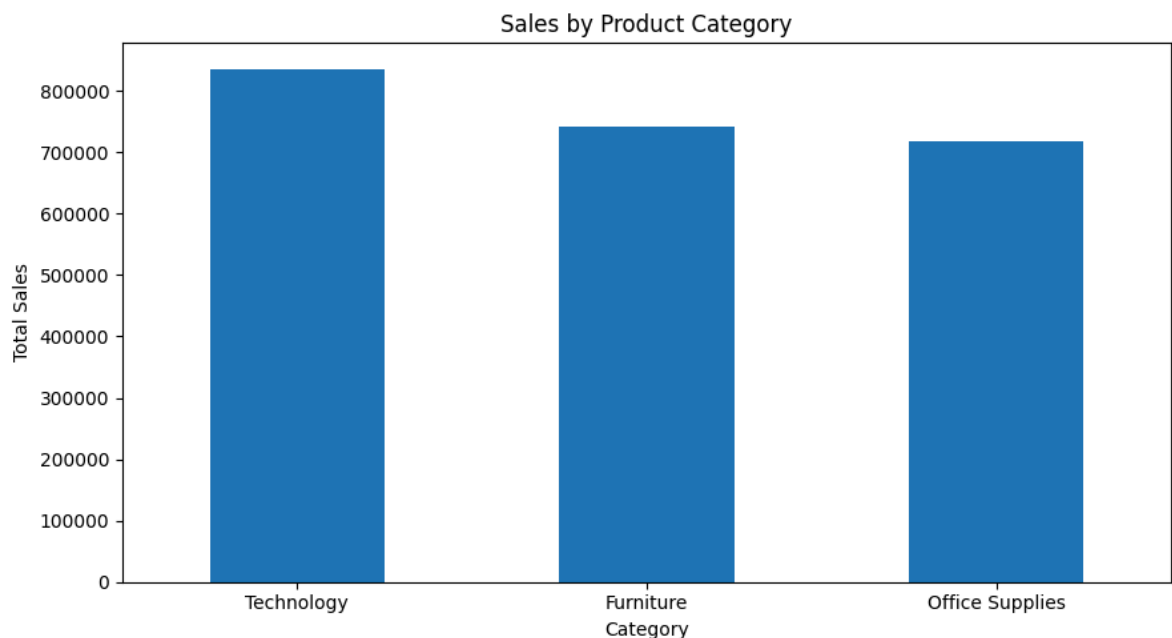
plt.figure(figsize=(9, 5))

category_sales.plot(kind='bar')

plt.title('Sales by Product Category')
plt.xlabel('Category')
plt.ylabel('Total Sales')
plt.xticks(rotation=0)

plt.tight_layout()
plt.show()

print(category_sales)
```



```
Category
Technology      836154.0330
Furniture        741999.7953
Office Supplies  719047.0320
Name: Sales, dtype: float64
```

✓ Sales by Region

```
region_sales = df.groupby('Region')['Sales'].sum().sort_values(ascending=False)

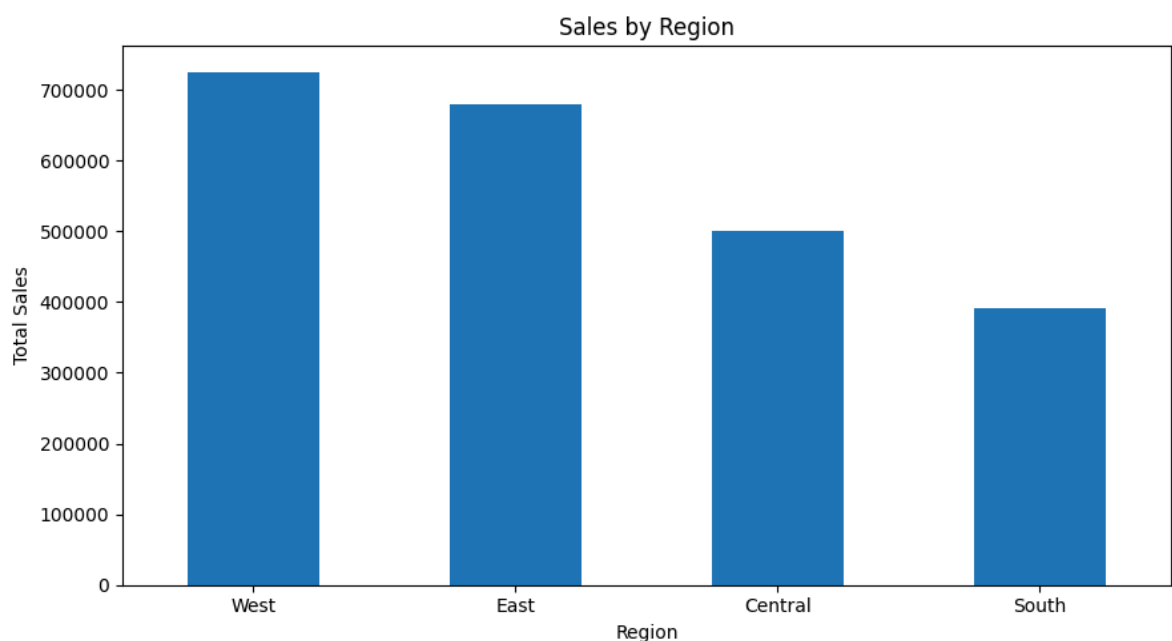
plt.figure(figsize=(9, 5))

region_sales.plot(kind='bar')

plt.title('Sales by Region')
plt.xlabel('Region')
plt.ylabel('Total Sales')
plt.xticks(rotation=0)

plt.tight_layout()
plt.show()

print(region_sales)
```



```
Region
West      725457.8245
East      678781.2400
Central   501239.8908
South     391721.9050
Name: Sales, dtype: float64
```

✓ Profit by Product Category

```
category_profit = df.groupby('Category')['Profit'].sum().sort_values(ascendi

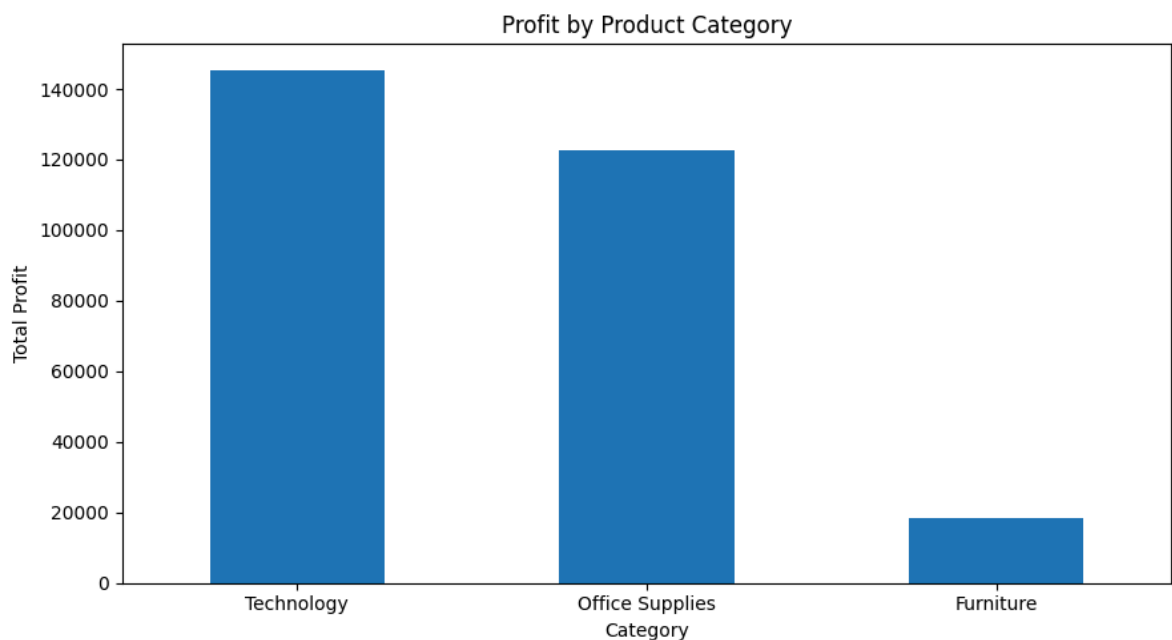
plt.figure(figsize=(9, 5))

category_profit.plot(kind='bar')

plt.title('Profit by Product Category')
plt.xlabel('Category')
plt.ylabel('Total Profit')
plt.xticks(rotation=0)

plt.tight_layout()
plt.show()

print(category_profit)
```

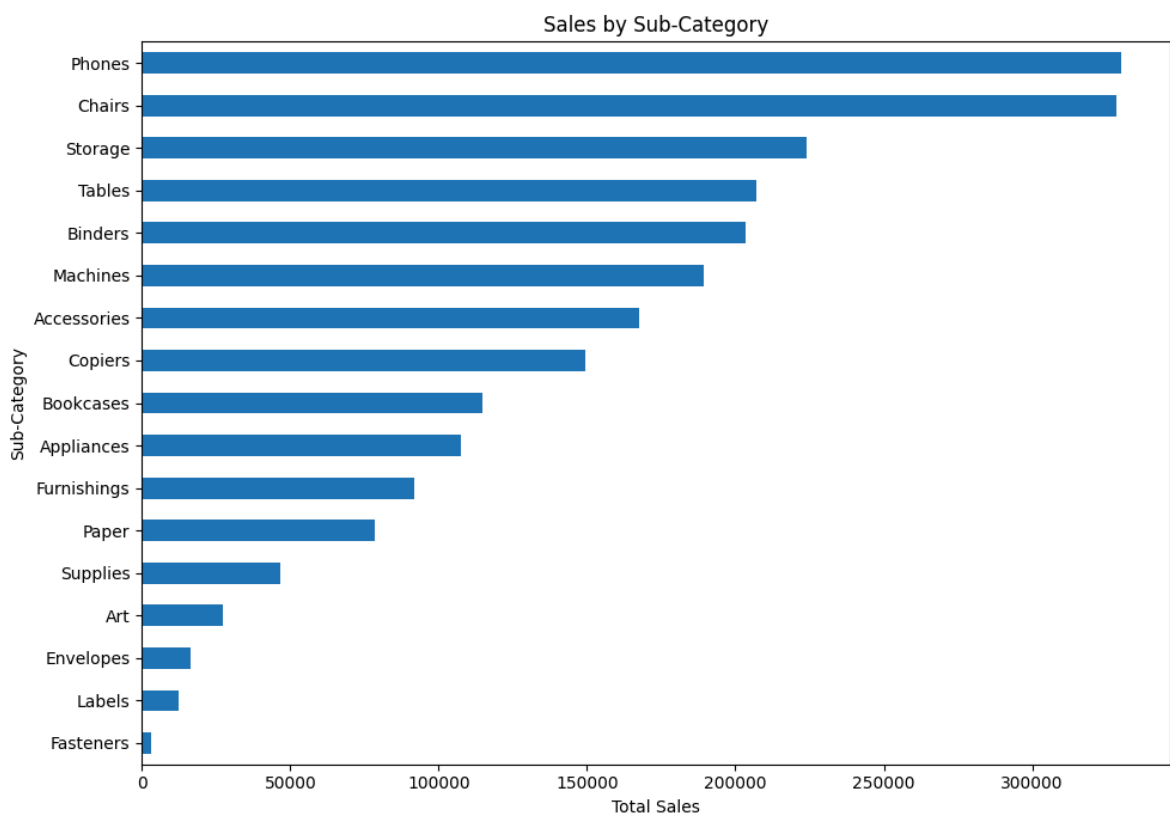


```
Category
Technology      145454.9481
Office Supplies  122490.8008
Furniture        18451.2728
Name: Profit, dtype: float64
```

✓ Sales by Sub-Category

```
subcategory_sales = (
    df.groupby('Sub-Category')['Sales']
    .sum()
```

```
.sort_values()  
)  
  
plt.figure(figsize=(10, 7))  
  
subcategory_sales.plot(kind='barh')  
  
plt.title('Sales by Sub-Category')  
plt.xlabel('Total Sales')  
plt.ylabel('Sub-Category')  
  
plt.tight_layout()  
plt.show()
```

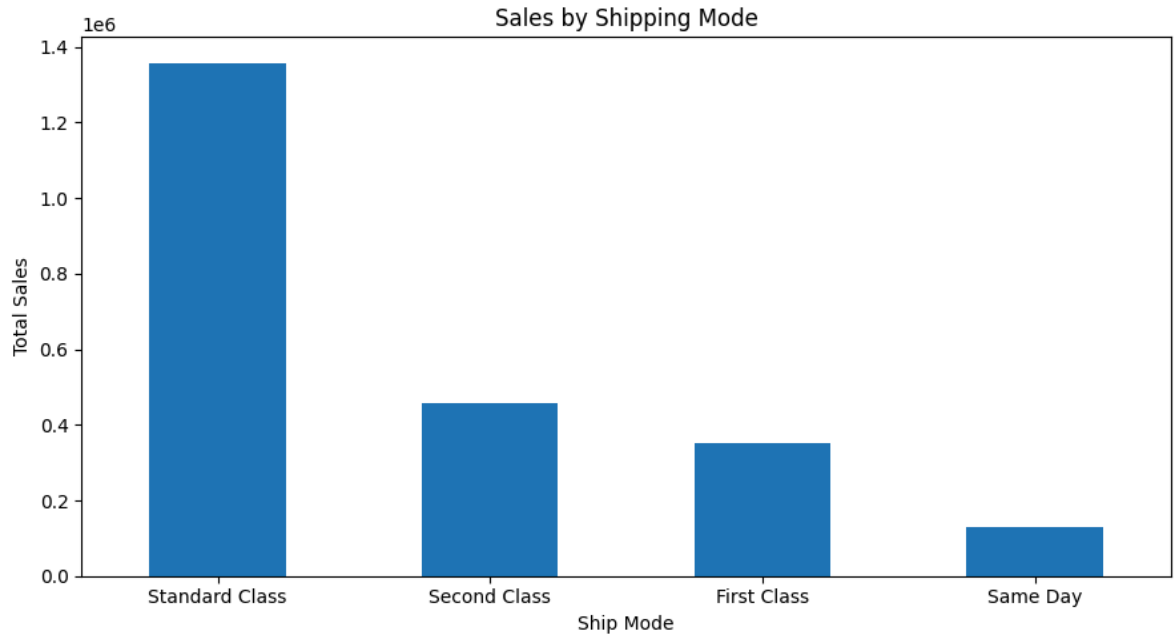


```
print("Top 5 Sub-Categories:")  
print(subcategory_sales.sort_values(ascending=False).head(5))
```

```
Top 5 Sub-Categories:  
Sub-Category  
Phones      330007.054  
Chairs      328449.103  
Storage     223843.608  
Tables      206965.532  
Binders     203412.733  
Name: Sales, dtype: float64
```

✓ Sales by Shipping Mode

```
ship_sales = df.groupby('Ship Mode')['Sales'].sum().sort_values(ascending=Fa  
  
plt.figure(figsize=(9, 5))  
  
ship_sales.plot(kind='bar')  
  
plt.title('Sales by Shipping Mode')  
plt.xlabel('Ship Mode')  
plt.ylabel('Total Sales')  
plt.xticks(rotation=0)  
  
plt.tight_layout()  
plt.show()  
  
print(ship_sales)
```



```
Ship Mode
Standard Class    1.358216e+06
Second Class     4.591936e+05
First Class       3.514284e+05
Same Day          1.283631e+05
Name: Sales, dtype: float64
```

✓ Discount vs Profit

```
plt.figure(figsize=(10, 6))

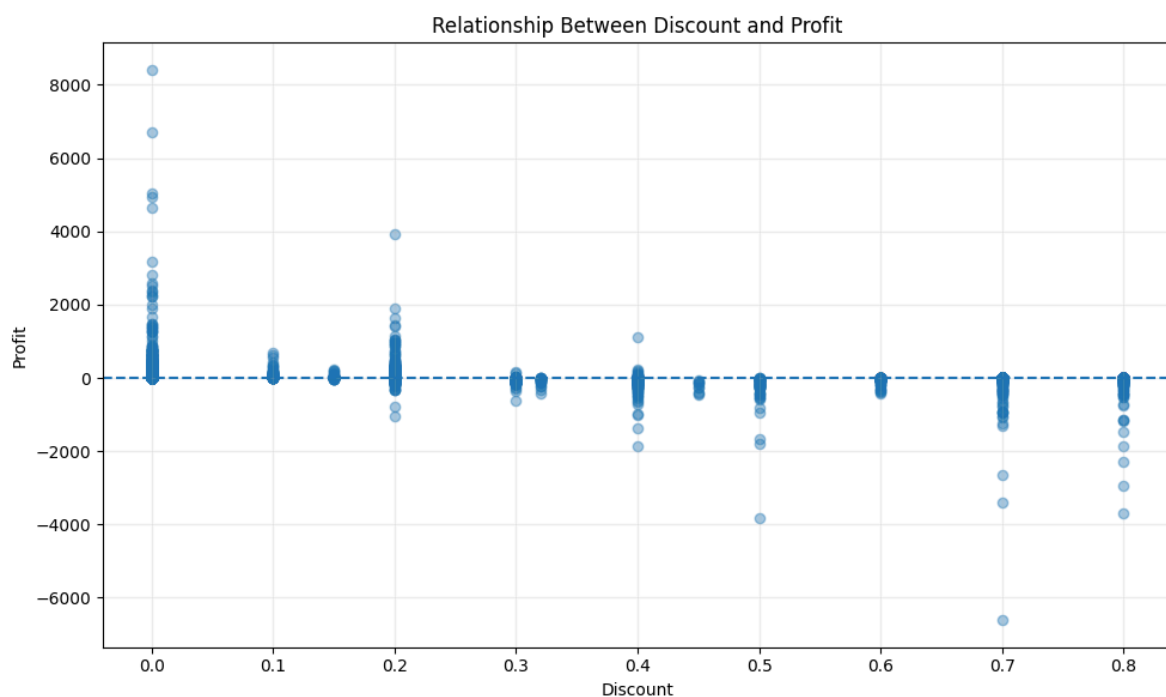
plt.scatter(
    df['Discount'],
    df['Profit'],
    alpha=0.4
)

plt.title('Relationship Between Discount and Profit')
plt.xlabel('Discount')
plt.ylabel('Profit')

plt.axhline(0, linestyle='--')

plt.grid(True, alpha=0.2)

plt.tight_layout()
plt.show()
```



```
total_sales = df['Sales'].sum()
total_profit = df['Profit'].sum()
total_quantity = df['Quantity'].sum()
```

```
print("===== SALES SUMMARY =====")
print(f"Total Sales      : ${total_sales:,.2f}")
print(f"Total Profit     : ${total_profit:,.2f}")
print(f"Total Quantity    : {total_quantity:,.0f}")
print(f"Profit Margin      : {(total_profit / total_sales) * 100:.2f}%")
```

```
===== SALES SUMMARY =====
Total Sales      : $2,297,200.86
Total Profit     : $286,397.02
Total Quantity   : 37,873
Profit Margin    : 12.47%
```

```
print("===== KEY FINDINGS =====")

print("\nHighest Sales Category:")
```